New Potential for Multimodal Connection: Exploring the Relationship between Taxi and Transit in New York City (NYC)

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ABSTRACT

Taxi trips have been somewhat neglected in transportation mobility and multimodal connection studies. The socio-demographic characteristics of taxi riders are often not fully revealed and the relationship between the taxi and fixed-route public transportation has not been sufficiently quantified. This research operationalizes the multifaceted relationship between the taxi and transit in an innovative way. We categorize taxi trips into three types: transit-competing, transit-complementing, and transit-extending trips, by examining the spatial relationship of taxi trips' origins/destinations and the locations of transit stations using a New York global positioning system taxi trip dataset. The distinct characteristics of the three types of taxi trips reflect the different market segmentations that taxis serve and the competing, complementary, and supportive nature of the relationship between the taxi and transit. We also explore the demographic characteristics of taxi riders and the result reveals the important role of taxis in providing mobility options to the economically or physically challenged population. Among our many findings are that transit-extending taxi trips have significantly shorter average trip lengths and larger proportions of people paying with cash than other trip types. These and other results point to important policy implications for improving the multimodal connection between the taxi and transit.

Keywords: Taxi, multimodal connection, first and last mile, public transportation, paratransit

INTRODUCTION

Taxis, a form of on-demand public transportation, have been neglected in most transportation mobility and multimodal connection studies (King, Peters, and Daus 2012). However, they have the potential to serve the travel demand unmet by other transportation modes and may facilitate multimodal connections of a transport system since they serve point-to-point, do not require car ownership, and sometimes support the usage of fixed-route public transportation. Only recently have emerging travel modes like app-based on-demand ride services (ride-sourcing) become prevalent and taxis seem to receive more attention. With the obvious similarity between the taxi and ride-sourcing, taxi trips are a good reference to further our understanding of ride-sourcing or other emerging on-demand ride services like shared autonomous vehicles as they become more established travel options.

Strategies to increase transit ridership and encourage mode shift have been widely studied and many of them are related to built environment factors or potential customers' ability to access transit (Ewing and Cervero 2001; Cervero and Kockelman 1997; Cervero 1994). Although transit ridership and people's mode choice are influenced by multiple factors, the limited coverage of transit service in many American cities discourages transit use. Emerging transportation technologies, such as ride-sourcing and autonomous vehicles, may bring about changes in people's travel behaviors and shift current travel mode shares. Though ride-sourcing has been criticized for competing with public transit, it also creates potential as a new way to access public transit (Rayle et al. 2016; Smart et al. 2015). Little research has examined the relationship between on-demand ride services and transit, making it difficult to understand how emerging travel modes can play a role in facilitating transit use in a multimodal context. In brief, understanding the relationship between the taxi and transit can provide insights, not only into the current multimodal connections of a transport system but also into the potential influence of emerging travel modes, like ride-sourcing and shared autonomous vehicles.

This research identifies the socio-demographic characteristics of taxi riders and operationalizes the relationship between the taxi and fixed-route mass transit. The study employs the New York-New Jersey-Connecticut (NY-NJ-CT) 2010/2011 Regional Household Travel Survey (RHTS) Data and the GPS taxi trip data of New York City (NYC). Based on the spatial relationship between taxi trips and fixed-route transit stations, we classify taxi trips into three types, namely transit-extending, transit-competing, and transit-complementing trips, and we identify distinct characteristics of those taxi trips and taxi riders. Through quantifying the competing, complementary, and supportive nature of the relationship between the taxi and transit, and by describing the notable characteristics of taxi riders, the paper confirms a paratransit-type role for taxis, facilitates better understanding of the multifaceted relationship between the taxi and transit, and provides implications for improving mobility and multimodal connectivity.

LITERATURE REVIEW

The taxi is a form of on-demand public transportation and sometimes serves as paratransit. Paratransit is defined as flexible curb-to-curb public transportation often specialized in serving people who are not able to ride fixed-route public transit (Cervero 1997). In general, paratransit can encourage modal shifts away from cars, enhance mobility especially in poor neighborhoods, and take on some of the peak demand of a transportation system (Cervero 1997). Key operational advantages of the taxi include its low capital requirements and its ability to serve a wide range of origin-destination pairs (Austin and Zegras 2011). However, compared to other travel modes, the taxi is often overlooked in research, which might be attributable to three reasons according to King,

Peters, and Daus (2012). First, unlike conventional transit, taxis are not directly subsidized, so there is little motivation for publicly funded research to model the cost-effectiveness of taxi investment. Second, taxi service is often viewed as a luxury good, even though data and studies have shown that taxis actually serve more low-income people than the rich (Altshuler, Womack, and Pucher 1980; King, Peters, and Daus 2012; Pucher and Renne 2003). Lastly, taxis do not have a great number of riders in most American cities because people more often rely on driving. The following review summarizes the findings from the existing literature on the demand for taxi service, the characteristics of taxi trips, and the relationship between the taxi and fixed-route transit.

The Demand for Taxi Service

Evidence from existing literature suggests that taxis play an important role in serving the unmet travel demand of the low-income population. The spatial distribution of taxi trips is found to have a significant correlation with that of transit access and low-income households (Austin and Zegras 2011; Kattan, de Barros, and Wirasinghe 2010; Schaller 2005). King and Saldarriaga (2016) found that taxi users who pay with cash are associated with unbanked and low-income households.

Past studies also noted that taxis serve the elderly and disabled people disproportionately (Rosenbloom 2003). Many areas in London subsidize a benefit called Taxicard that can be used to pay for taxi rides by the elderly and disabled people (Schmöcker et al. 2008; Schmöcker et al. 2005). Schmöcker et al. (2005) found that elderly and disabled people generally have fewer and shorter trips traveled, and the possession of a Taxicard is associated with more trips generated by both the elderly and disabled groups. A study in Nigeria showed that the elderly use taxis for 20% of their travel which is a significantly larger proportion compared to other age cohorts (Ipingbemi 2010). Taxis are also considered adept at serving special trip purposes, such as trips to/from large transportation hubs (railway station or airport) or large shopping venues (Gupta, Vovsha, and Donnelly 2008; Schaller 2005).

Characteristics of Taxi Trips

Taxi trips are different from both transit and automobile trips. The spatial and temporal distribution of taxi trips varies by time of day, the day of the week, season, weather, holidays, urban environment, etc. (Chang, Tai, and Hsu 2009; Kamga, Yazici, and Singhal 2013; Phithakkitnukoon et al. 2010). The great variation makes it difficult to characterize taxi trips or forecast the demand for taxi service. Taxi trips are also characterized by trip pattern asymmetry (King, Peters, and Daus 2012; Lacombe and Morency 2016; Qian, Zhan, and Ukkusuri 2015), seasonality (Hochmair 2016), high density of specific trip purposes (Espín Noboa et al. 2016), and irregular users (Christoforou et al. 2012).

Several existing studies focus on modeling taxi trip generation with socio-demographic and built environment factors. Lacombe and Morency (2016) modeled taxi trip generation in Montreal, Canada and noted that variables of income, age, transit access time, and number of parking spots are associated with both taxi pick-ups and drop-offs, while job types affect only drop-offs. Yang and Gonzales (2014) discovered that population, median age of the origin or destination district, education level, median household income, and job count significantly correlate with the number of taxi trip pick-ups/drop-offs at the census tract level. Qian, Zhan, and Ukkusuri (2015) mentioned that land use is an important factor associated with the number of taxi trips starting or ending in an area. Qian and Ukkusuri (2015) found that urban form factors,

represented by residential density, road density, and subway accessibility, are also significantly correlated with taxi trip generation.

The Relationship between Taxi and Transit

Early studies have elaborated the relationship between the taxi and transit despite an initial lack of empirical evidence. Gilbert and Samuels (1982) see taxis as one solution for providing satisfactory mobility to certain urban areas where conventional transit is less cost-effective. They also argue that public transit programs and subsidies should incorporate taxis (Austin and Zegras 2011). Webster, Wiener, and Wells (1974) advocate for integrating taxi services and mass transit to provide better intra-city public transportation.

Recently, there have been more quantitative studies on the relationship between taxi trips and transit trips, partly because some empirical taxi trip data have become available. For instance, Hochmair (2016) examined the relationship between taxi trips and other explanatory variables, including the availability of subway and bus, and identified the number of subway/train stations to be positively associated with taxi trip counts in NYC. Hochmair (2016) also found that the number of bus stops is negatively associated with taxi trip counts, which might indicate that taxis compete with buses in well-served areas. By examining the relationship between taxi trips and transit accessibility in Boston, Austin and Zegras (2011) concluded that taxis act as both a substitute and complement to mass transit.

In addition to having either a competing or complementary nature vis-à-vis fixed-route public transit, taxis can serve the first and last mile of transit trips. The quality of transport networks not only depends on individual links and nodes but also on how these nodes and links function in a multimodal context (Keijer and Rietveld 2000). Kattan, de Barros, and Wirasinghe (2010) noted that the high correlation between public transit ridership and taxi trips could be explained by the direct demand for taxi service from major transit stations. Multimodal trips have historically been underrepresented in travel surveys, which leads to a lack of effective empirical data to understand multimodal travel behavior (Clifton and Muhs 2012). Even though on-demand ride services possess great potential as an effective access mode to public transportation, the role of taxi service as an access mode to transit requires greater study.

Gaps in the Literature and Motivation

This review covers most relevant literature on taxi trips and depicts a general picture of the demand and travel characteristics of taxi service, while still leaving some questions open. The taxi plays an important role in providing mobility to low-income, elderly, the disabled populations, and in serving irregular and one-way trips. Assorted socio-demographic, urban form, economic, and temporal factors are associated with variation in the demand for taxi trips, but consensus has yet to be reached due to the limited number of studies that combine different locations, scales, and data sources. Moreover, the relationship between the taxi and mass transit has not been specified. A primary reason is the lack of comparable data sources for taxis and transit, but more importantly, the relationship between the taxi and transit is multifaceted and difficult to quantify: in different situations, the taxi may replace, complement, or serve the first/last miles of transit trips.

This research investigates the socio-demographic characteristics of taxi riders and operationalizes the relationship between the taxi and transit by distinguishing among three types of taxi trips (i.e. transit-complementing, transit-competing, and transit-extending taxi trips) with GPS taxi data in NYC and the NY-NJ-CT 2010/2011 RHTS data. By describing the socio-demographic characteristics of taxi riders and by comparing the characteristics of the three distinct types of taxi trips, the research furthers our understanding about the travel demand that taxis serve

and their complex relationship with fixed-route public transportation. The research focuses on specifying the synergetic relationship between the taxi and transit to derive policy implications for improving the multimodal connectivity.

DATA AND METHODOLOGY

Identifying the Characteristics of Taxi Riders in the New York Metropolitan Area

The socio-demographic characteristics of taxi riders remain somewhat obscure, partly due to the limited attention that taxi trips have received in previous research. Information of taxi trips and taxi riders is often missing from household travel surveys and even if the taxi is included in a survey, there may not be sufficient number of taxi users to allow effective statistical analysis. The New York metropolitan area is an exception: even though the taxi has a very small mode share, it represents a substantial number of trips. The 2010/2011 RHTS data, covering 28 counties of the NY-NJ-CT metropolitan area, shows that there are 1,458 trips made by taxi out of the total 143,926 trips. We compared the characteristics of taxi riders with travelers using other modes to extract the socio-demographic profiles of taxi riders.

Classifying the Three Types of Taxi Trips

To quantify the relationship between taxis and fixed-route public transportation, we categorized taxi trips into three distinct types. The three types are denoted as transit-extending, transit-competing, and transit-complementing trips and are defined as follows:

(1) Transit-extending taxi trips provide connectivity to/from subway or train stations.
(2) Transit-competing taxi trips are taxi trips that could have been made by taking subway or train.
(3) Transit-complementing taxi trips serve routes and operate at times at which the subway and train do not.

The classification process utilized the NYC GPS taxi trip data that is available on the website of the NYC Taxi & Limousine Commission. The dataset contains information about every taxi trip's pick-up location, drop-off location, travel time, cost, number of passengers, trip length, etc., and includes all taxi trips from 2009 to 2016. We used a four-week section of taxi trip data in 2011 as our data pool, including the week of January 24th - 30th, April 4th - 10th, July 18th - 24th, and October 17th - 23rd, 2011. The four weeks were chosen because they avoid major holidays and imply sufficient seasonal variation. We chose to use the 2011 data because ride-sourcing, like Uber, which is believed to replace taxi trips, had not yet entered the market, so the data reflects taxi demand more completely. Also, the 'green taxi' that is only allowed to pick up passengers from outer boroughs was not on the market in 2011 so there was no restriction on the pick-up locations of taxi service. Next, we extracted a random sample of one million trips out of this four-week data pool that contains over nine million taxi trips to allow efficient computation for the classification and the regression analysis (the next step). The final dataset used for the analysis has 983,053 taxi trips, after removing observations with missing or outlier values (e.g. trips longer than 100 miles). It accounts for approximately ten percent of the size of the original data pool and should be a valid sample size.

Taxis were allowed to pick up passengers anywhere in the five boroughs of New York in 2011. Many of the taxi trips in our data pool cover the eastern part of New Jersey, like Jersey City, North Bergen, and Bayonne. Therefore, the study area was set to a six-county area including New York, Bronx, Queens, and Kings in New York State, and Hudson and Bergen Counties in New

Jersey, as shown in Figure 1. More than 99% of the taxi trips fall into this six-county area. The subway and train lines included in our analysis are also shown in Figure 1, and all 1,326 subway or train stations along these lines that fall into the six-county study area were included in the classification process. Among the 1,326 stations, 899 are New York subway stations, 341 are along Long Island Railroad lines, 15 are along Metro-North Railroad lines, and 71 are along New Jersey light rail lines. The lines and stations of the subway and trains constitute the 'transit system' in the classification of taxi trips.

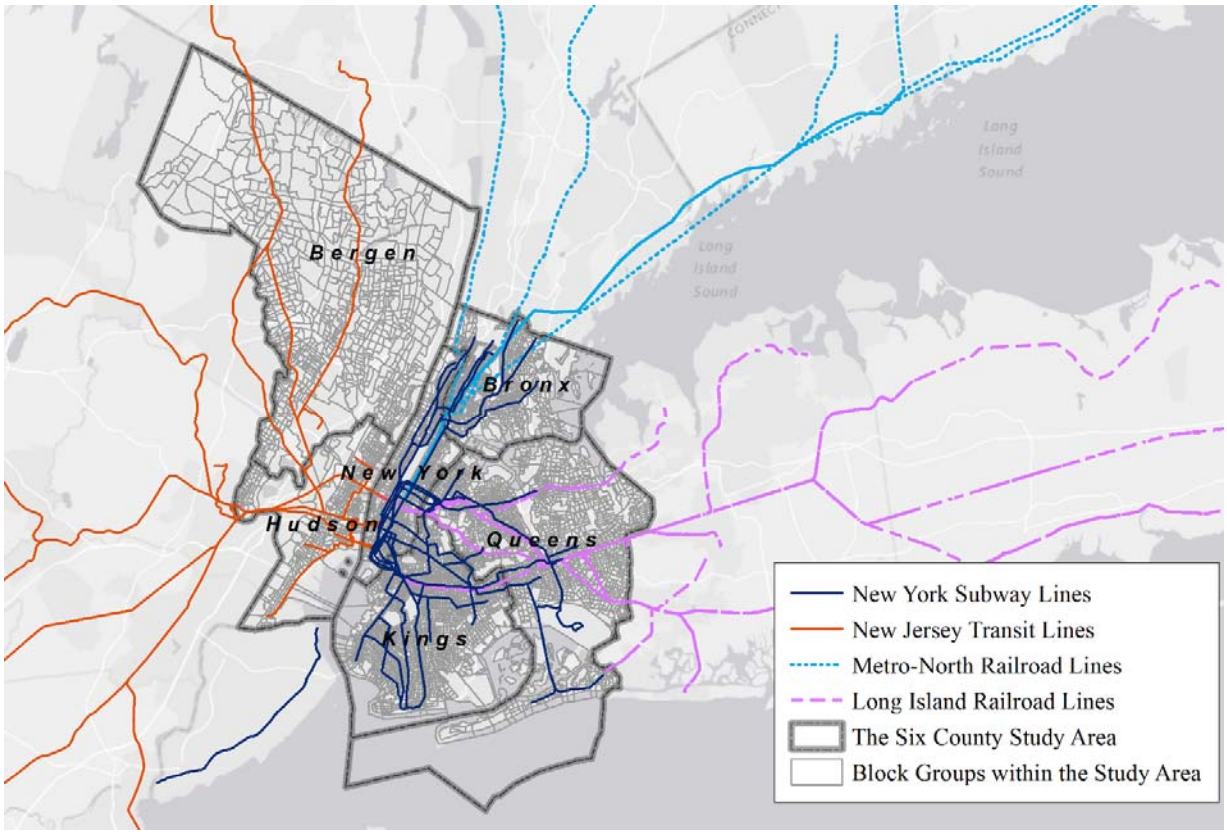

Figure 1. The Study Area and Transit Lines. Source: Shapefiles were downloaded from the official website of ArcGIS / ESRI

Bus stops were not included in the classification. There are over 22 thousand bus stops in our study area. Including bus stops into the classification would have classified more than 90% of the taxi trips as 'transit-competing,' even though this classification does not contribute much to our knowledge about the nature of the relationship between the taxi and transit. Although the majority of taxi trips could have been made by bus because of the large extent of the bus system, a trip that requires multiple transfers (such as transfer from subway to bus) or requires long out-of-vehicle wait time is not comparable to a trip made exclusively by subway, train, or taxi. By contrast, we found that all the origin/destination pairs using the subway require no more than two transfers, and the subway runs 24 hours a day with short headways. Therefore, the classification incorporated only subway and train lines. The lower level of service of buses makes them difficult to compare with trains, subways, or taxis for trip classification purposes.

Figure 2 is a conceptual illustration of the three types of taxi trips that we classified. Generally, the classification hinges on whether a taxi trip could have been made by taking transit, serves a route or time that the transit system does not serve, or is likely serving as an access/egress mode to the transit. We used Python scripting to compute the relationship between the transit system (subway and train) and the origin/destination of each of the 983,053 taxi trips. We also considered the operating hours at all the transit stations to perform the classification. Figure 3 shows the stepwise logic of the classification and the detailed classification process is illustrated below.

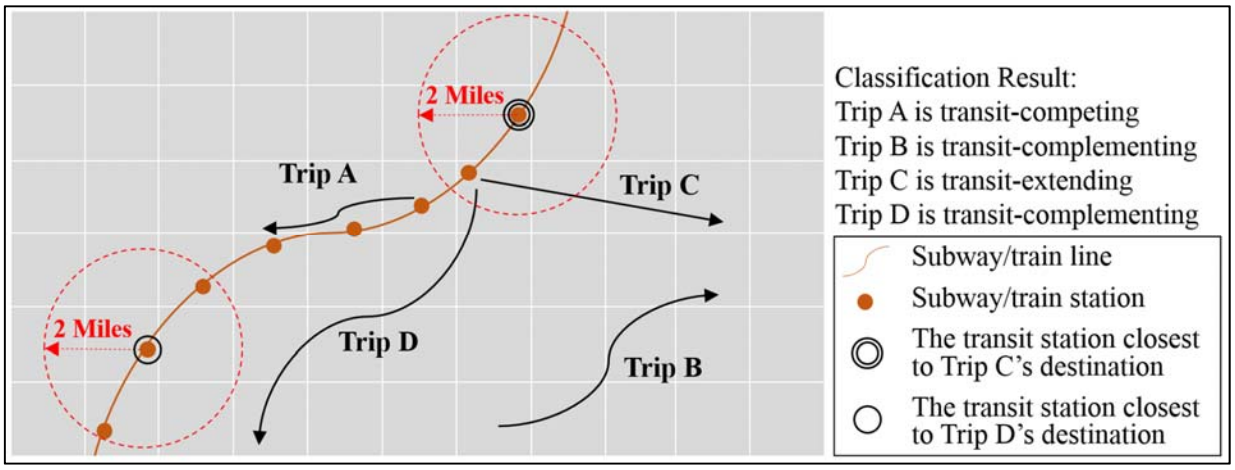

Figure 2. Conceptual Example of Classifying the Three Types of Taxi Trips

Classifying Transit-extending taxi trips

Transit-extending taxi trips are defined as taxi trips that connect to/from transit. Classifying transit-extending taxi trips requires selecting trips where taxis most likely serve as the access/egress mode of transit. We denote the taxi trips that connect to transit as 'transit-accessing' and the ones connecting from transit as 'transit-egressing'. Transit-extending trips were selected from the taxi trip data pool according to three sequential rules: (1) only one trip end (either origin or destination) has a transit station within 0.125 miles, and the trip took place during the stations' operating hours; (2) the other trip end does not have a transit station within 0.25 miles; and (3) the transit station associated with one end of the trip is within two miles of the transit station nearest the other end of the taxi trip. This third step excludes the transit stations counter to the direction of travel. More specifically, if a taxi trip's origin is within 0.125 miles of a transit station and its destination is over 0.25 miles from a transit station, the trip may not necessarily be classified as transit-extending. We need to make sure whether the transit station within 0.125 miles of the trip origin is also within a two-mile buffer of the transit station nearest the trip destination to then classify the trip as transit-extending, and more specifically, transit-egressing. A transit extending trip can also flip the origin's and destination's location vis-à-vis transit stations, which will make the trip classified as transit-accessing.

For example, as shown in Figure 2, Trip C starts within 0.125 miles of a transit station that is located within two miles of the station nearest the trip destination. Also, the destination is not within 0.25 miles of a transit station, so Trip C is classified as transit-extending. More specifically, it is a taxi trip that connects transit to unserved regions and is a transit-egressing trip. In contrast, Trip D starts within 0.125 miles of a transit station, but that station does not fall into the two-mile

buffer of the station nearest the destination of Trip D. Therefore, Trip D is classified as transit-complementing rather than transit-extending. We can easily understand that if the rider of Trip D intends to take a taxi to leave a transit station, he/she would have chosen to get off at a station that is closer to his/her final destination, such as the three stations falling into the two-mile buffer of the station nearest the trip destination. In Figure 3, the logic path of (1) => (3) => (4) forms the classification process of 'transit-extending' taxi trip that connects from transit (transit-egressing). The logic path of (6) => (7) => (9) => (11) forms the classification process of "transit-extending" taxi trip that connects to transit (transit-accessing). The third filtering step was implemented by examining the two larger diamond boxes in Figure 3.

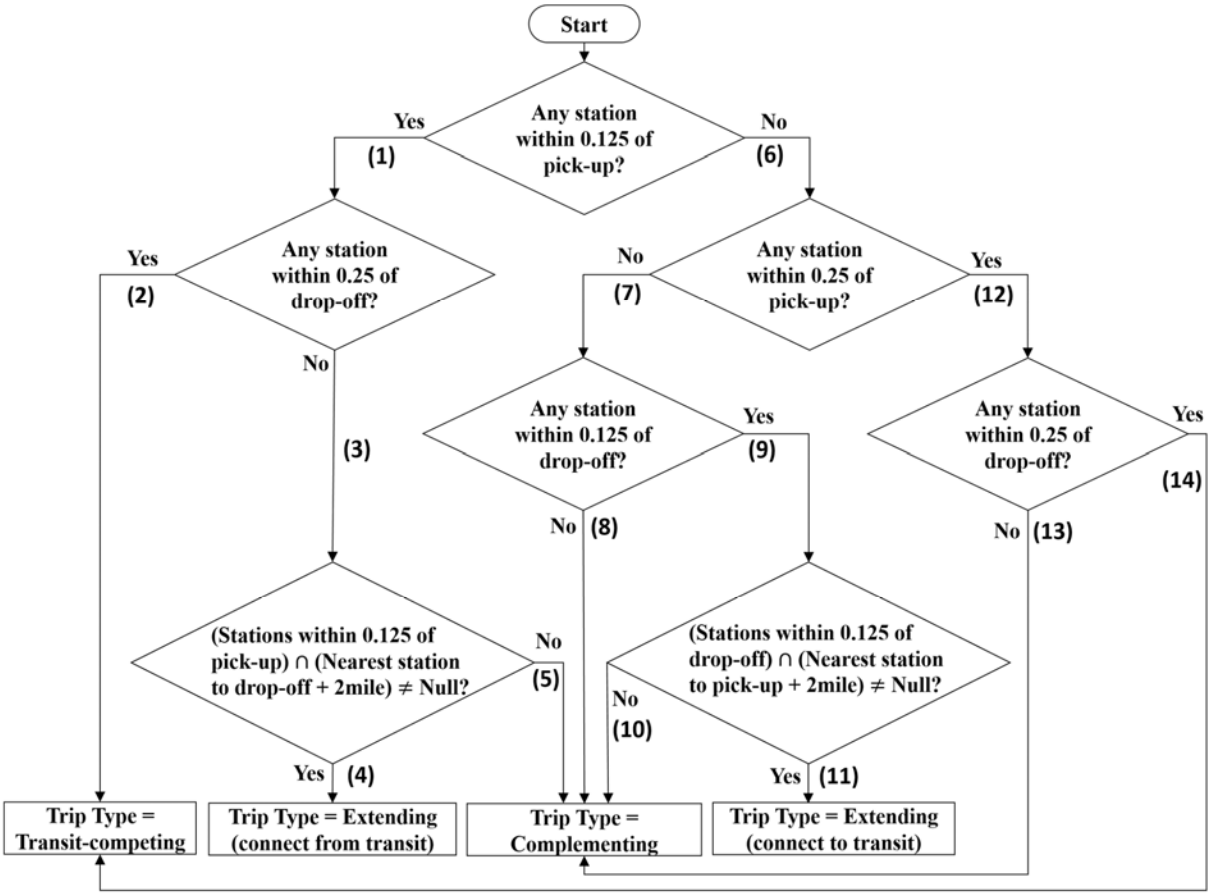

Figure 3. Logical Flow Chart of Selecting Transit-Extending, Transit-Competing, and Transit-Complementing Taxi Trips (Numbers in the Parentheses Refer to the Logical Links Discussed)

Classifying Transit-competing taxi trips

Transit-competing taxi trips are defined as trips that could have been achieved by taking transit. Transit-competing trips were classified by selecting the trips that have at least one subway/train station within 0.25 miles of both trip ends. Also, the trips had to take place when the corresponding transit service was operating. This rule of selecting transit-competing taxi trips guarantees that the same trip could have been achieved by taking transit with walking no more than 0.5 miles between the trip origin and destination. Trip A in Figure 2 is an intuitional example of such trips, as it has transit stations within 0.25 miles from both the trip origin and destination.

Accordingly, in Figure (3), the logic path of (1) => (2) and the logic path of (6) => (12) => (14) form the classification processes of 'transit-competing' taxi trips.

Classifying Transit-complementing taxi trips

Transit-complementing taxi trips are defined as taxi trips that serve routes (origin to destination) or time that the transit system does not serve. The trips were selected by excluding the previous two trip types. Trip B in Figure 2 is an example of such trips, as there is no transit station near the trip origin or destination. Accordingly, in Figure 3, four logic paths classify transit-complementing taxi trips, including the path that goes through links (1) => (3) => (5); links (6) => (7) => (8); links (6) => (7) => (9) => (10); and links (6) => (12) => (13).

As described, the classification of the taxi trips is based on examining the geographic locations of taxi trips' origins/destinations vis-à-vis transit stations. The classification based on such spatial examination allows us to quantify to what degree the taxi provides access to transit ('transit-extending'), replaces transit trips ('transit-competing'), and serves trips not feasible by transit ('transit-complementing'). The classification method has its limitations. First, it cannot be validated with currently available data; and second, the classification cannot fully encapsulate the nature of transit competition or complementarity. For example, if a traveler normally uses transit but takes a taxi when it is raining, the taxi trip will be classified as transit-competing even though its true nature is more complementary. However, the classification of the three types of taxi trips allows us to specify how taxis are generally utilized as a function of the availability of fixed-route transit and to further understand how to maximize the synergetic relationship while minimizing the conflict between the taxi and transit.

Binary Logit Models

Two binary logit regression models were developed to further examine the characteristics of the three types of taxi trips. The first binary logit model predicts whether a taxi trip is transit-competing, and the second model predicts whether a taxi trip is transit-extending, given trip characteristics, weather conditions, and socio-economic and built environment factors associated with trip origins and destinations. The variables included in the two models, their data sources, and geographic scales are summarized in Table 1. Both models were developed using the categorized taxi trip observations from the previous classification. The linkage of different types of taxi trips with the economic, built environment, and trip-level factors allows us to further understand taxi travel demand and the areas that each of the three types of taxi trips serves.

Table 1. Variables Tested in the Binary Logit Regression Models

Variable Categories	Variable Name	Explanation	Data Source
Dependent variable for Model 1	Transit-Competing	A dummy indicating whether it is a transit-competing taxi trip	NYC GPS taxi trip data
Dependent variable for Model 2	Transit-Extending	A dummy indicating whether it is a transit-extending taxi trip	
Trip variables	Passenger	Number of passengers of the ride	NYC GPS taxi trip data
	Log distance	The log transformed trip length (in miles)	
	Payment cash	A dummy indicating whether the trip was paid by cash (vs. by card)	
	Total cost	Total cost of the trip (including fare, tip, and toll)	

	Morning peak	A dummy indicating whether the trip was between 6 am to 9 am	
	Evening peak	A dummy indicating whether the trip was between 4 pm to 7 pm	
	Late night	A dummy indicating whether the trip was between 9 pm and 4 am	
Weather variables	Precipitation	The hourly amount of precipitation when the trip took place (inch)	Weather Underground 2011 data
	Temperature	The temperature at that time (by hour) when the trip took place (°F)	
	Rain dummy	A dummy indicating whether there was rain in the hour when the trip took place	
Land use variables (at both origin and destination of the trip)	Residential	A dummy variable of low to medium density residential land use ("one and two family building" for New York; "Residential, single unit/low density/medium density" for New Jersey)	New York MapPLUTO land use database and New Jersey Departments of Environmental Protection land use and cover data are merged; Taxi trips' origin and destination are joined to the land use category that they fall into
	High residential	A dummy of high-density residential land use ("multifamily building" for New York; "Residential, high density or multiple dwelling" for New Jersey)	
	Mixed-use	A dummy of mixed use of residential and commercial	
	Commercial	A dummy of commercial land use	
	Industrial	A dummy of industrial land use	
	Other	A dummy of other land use, including vacant, open space, transportation, etc.	
Block group level variables (at both origin and destination of the trip)	Poverty rate	Percent of families below the poverty line	ACS 5-year estimate 2009 -2013
	Median housing value	Median housing value (in 2013 dollar)	
	Unemployment rate	Percent of unemployed population	
	Percent low-income jobs	Percent of low-income jobs (with earnings $1250/month or less)	LEHD data in 2011
	Percent high-income jobs	Percent of high-income jobs (with earnings greater than $3333/month)	
Grid level variables (at both origin and destination of the trip)	Employment density	Number of jobs per square mile	Calculated from LEHD 2010 data and New York and New Jersey roadway GIS data
	Population density	Number of persons per square mile	
	Intersection density	Number of 3-way and 4-way intersections per square mile	
	Employment-population balance	Ratio of jobs to persons	
	Bus stop density	Number of bus stops per square kilometer	Calculated from New York and New Jersey transit GIS data
	Bus line density	Miles of bus lines per square kilometer	

RESULTS AND ANALYSIS

Socio-Demographic Characteristics of Taxi Riders

According to the 2010/2011 RHTS data, the taxi has a mode share of about 1.01% in the 28 counties of the NY-NJ-CT metropolitan area. We developed Chi-squared tests and t-tests to compare the characteristics of taxi riders and travelers who choose other modes and Table 2 summarizes notable traits. Overall, taxi riders consist of larger proportions of female, disabled,

low-income, unemployed, and elderly/retired people, with lower average household vehicle ownership. Taxi riders' demographic characteristics are vastly different from drivers and also differ from transit riders, walkers, and bikers. For example, taxis have a much larger proportion of female riders while all other travel modes have similar gender use patterns. Regarding trip purpose, taxis disproportionately serve maintenance trips that are trips made for personal services, appointments, and shopping needed by the individual or household. This partly explains why there is higher proportion of female riders, as previous studies have suggested the linkage between women and household-related trips and women's lower car use compared with males in a household (Best and Lanzendorf 2005). Approximately 27% of taxi riders have a disability, which is significantly higher than the disability rate of travelers using any other mode. Taxis also serve a significantly larger portion of low-income and unemployed travelers, and taxi riders have the lowest average household vehicle ownership. About 14.8% of taxi riders are above 65 years old, and about 19.6% are retired, which is consistent with previous findings that taxis disproportionately serve elderly segments of the population. In sum, 899 out of the 1,458 taxi trips (59.5%) are made by people who are either disabled, low-income, elderly, retired or unemployed, which indicates the significant role that taxis play in providing mobility to the physically and economically disadvantaged population.

Table 2. Outstanding Socio-Demographic Characteristics of Taxi Riders and Taxi Trip Purposes

Demographic or Trip Factors		**Taxi**	**Chi-Squared Test** (see note)	**Drive**	**Transit**	**Bike/walk**	**All**
Gender	% of Female	61.8%	***	54.4%	52.1%	54.1%	53.9%
Disability	% of Disabled	27.0%	***	2.8%	5.6%	5.4%	3.7%
HH Vehicle Ownership	Average	0.6	*** (T-test)	2.2	0.8	0.9	1.8
Income Levels	% Low income (< 50,000)	51.8%	***	20.4%	40.3%	41.6%	25.8%
	% Med income (50,000 - 150,000)	29.0%		59.1%	41.7%	41.4%	54.4%
	% High income (> 150,000)	19.2%		20.6%	18.0%	17.0%	19.8%
Age	% < 16	4.0%	***	44.6%	5.4%	18.8%	14.6%
	% 16 - 64	81.1%		44.7%	86.6%	72.0%	74.8%
	% > 65	15.0%		10.7%	8.0%	9.2%	10.6%
Life Cycle	% Employed	54.7%	***	60.1%	70.1%	48.2%	57.6%
	% Unemployed	13.5%		5.3%	7.5%	9.8%	6.1%
	% Homemaker	5.8%		5.4%	2.4%	5.1%	4.9%
	% Adult student	1.2%		1.9%	3.5%	3.2%	2.2%
	% Retired	19.6%		12.4%	8.0%	11.3%	11.4%
	% Other	5.3%		14.9%	8.4%	22.4%	17.8%
Trip Purpose	% Work/School	41.0%	***	43.7%	68.8%	41.3%	47.9%
	% Home	1.4%		0.9%	0.8%	1.6%	0.9%
	% Recreational	16.2%		22.0%	10.7%	24.1%	20.3%
	% Maintenance	29.2%		11.2%	12.0%	12.0%	11.1%
	% Airport	1.2%		0.0%	0.0%	0.0%	0.1%
	% Other	10.9%		22.2%	7.7%	21.0%	19.8%

Note: Chi-squared tests were developed by comparing taxi trips vs. non-taxi trips for all categorical variables and a t-test (assuming different variances of two samples) was developed for taxi trips vs. non-taxi trips regarding household vehicle ownership. '***' indicates that we reject the null hypotheses (taxi trips are the same as non-taxi trips regarding that variable) at the 99.9% confidence level.

The Relationship between Taxi and Transit

We found that out of the 983,053 taxi trips, about 58.54% are transit-competing, 33.82% are transit-complementing, and about 7.64% are transit-extending. It is fair to say that the taxi has a multifaceted relationship with fixed-route mass transit. There is great variation of trip characteristics across the three trip types, as shown in Table 3. Trip distance is an important characteristic that distinguishes the three taxi trip types. The average trip length of transit-extending trips is only about 1.2 miles, compared to 2.1 miles of transit-competing trips and 4.1 miles of transit-complementing trips. Regarding payment method, transit-extending trips have a much lower rate (39.6%) of paying with cards, compared to 43.6% of transit-competing and 46.4% of transit-complementing trips respectively. There is not a great variation in the number of passengers across the three types. More details of trip characteristics are discussed below.

Table 3. Summary Statistics of Trip Characteristics of the Three Types of Taxi Trips

		Transit-Competing Trips	Transit-Complementing Trips	Transit-Extending Trips
No. of Observations		575486 (58.54%)	332531 (33.82%)	75065 (7.64%)
No. of Passengers	mean	1.66	1.66	1.64
	sd	1.25	1.25	1.24
	min	1.00	1.00	1.00
	max	6.00	6.00	6.00
Trip distance	mean	2.09	4.14	1.19
	sd	1.81	4.36	0.63
	min	0.10	0.10	0.10
	max	81.00	82.70	41.00
Total cost	mean	10.32	15.85	7.82
	sd	5.29	12.16	2.63
	min	2.50	2.50	2.50
	max	192.90	230.00	106.20
Payment method*	Card	250801 (43.6%)	154241 (46.4%)	29724 (39.6%)
	Cash	324685 (56.4%)	178290 (53.6%)	45341 (60.4%)

*Categorical variable

Trip Length

The average trip lengths of the three types of taxi trips vary significantly and the distributions are shown in Figure 4. Overall, the vast majority (about 95.6%) of all taxi trips are shorter than 10 miles. For the transit-competing taxi trips, about 75% are shorter than 2.6 miles, and the average is about 2.1 miles, indicating that taxi service might be cheap and fast enough over short travel distances to replace transit. Transit-complementing trips present a longer tail towards 20 miles of trip distance, reflecting the larger variation in trip length. The average length of transit-complementing trips is about 4.1 miles, which is statistically longer than the other two types of taxi trips, based on the t-test. Transit-extending trips have the shortest average trip length of about 1.2 miles, and 90.6% of all the transit-extending trips are between 0.5 miles and 2 miles. This implies that most of the trips taken to access transit stations in NYC are shorter than 2 miles and

beyond 0.5 miles, which is the commonly accepted walkable distance for accessing transit. The substantial difference in average trip lengths across the three types of taxi trips suggests that taxis are serving very different travel demands that may either extend, compete with, or complement transit services.

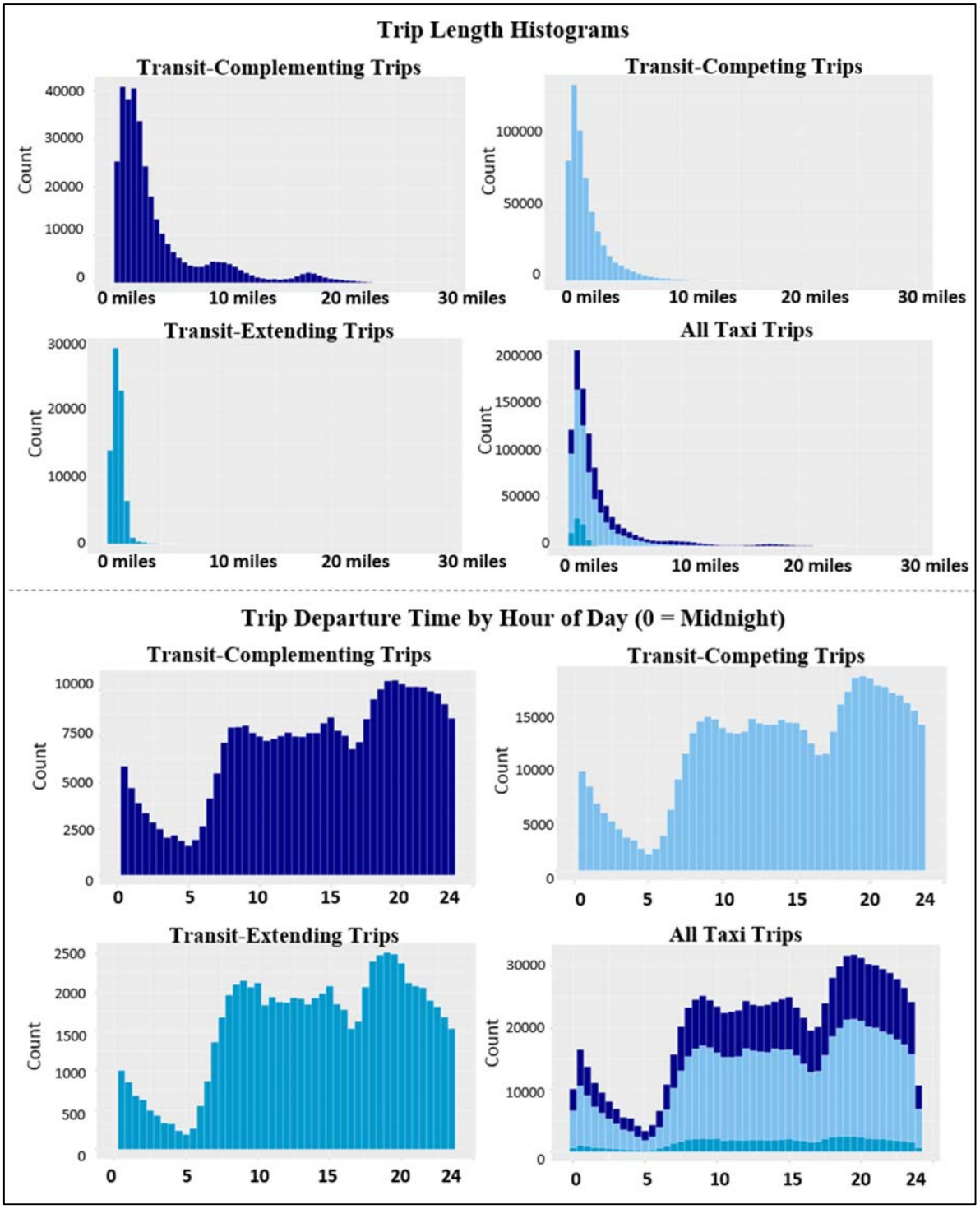

Figure 4. Trip Length and Time of Day Histograms of All Taxi Trips and the Three Categories

Time of Day

Figure 4 also presents the distributions of the departure times of taxi trips. The taxi trips in the study area generally occur during the daytime between 7 a.m. and midnight, peaking around 5 to 8 p.m. and bottoming out in the early morning around 4 to 5 a.m. None of the three types of taxi trips shows very distinct time-of-day patterns compared to the overall distribution, except that the transit-extending trips present a noticeable drop in the number of trips after 8 p.m., which might be due to travelers' reluctance to take transit late at night even though the subway still operates.

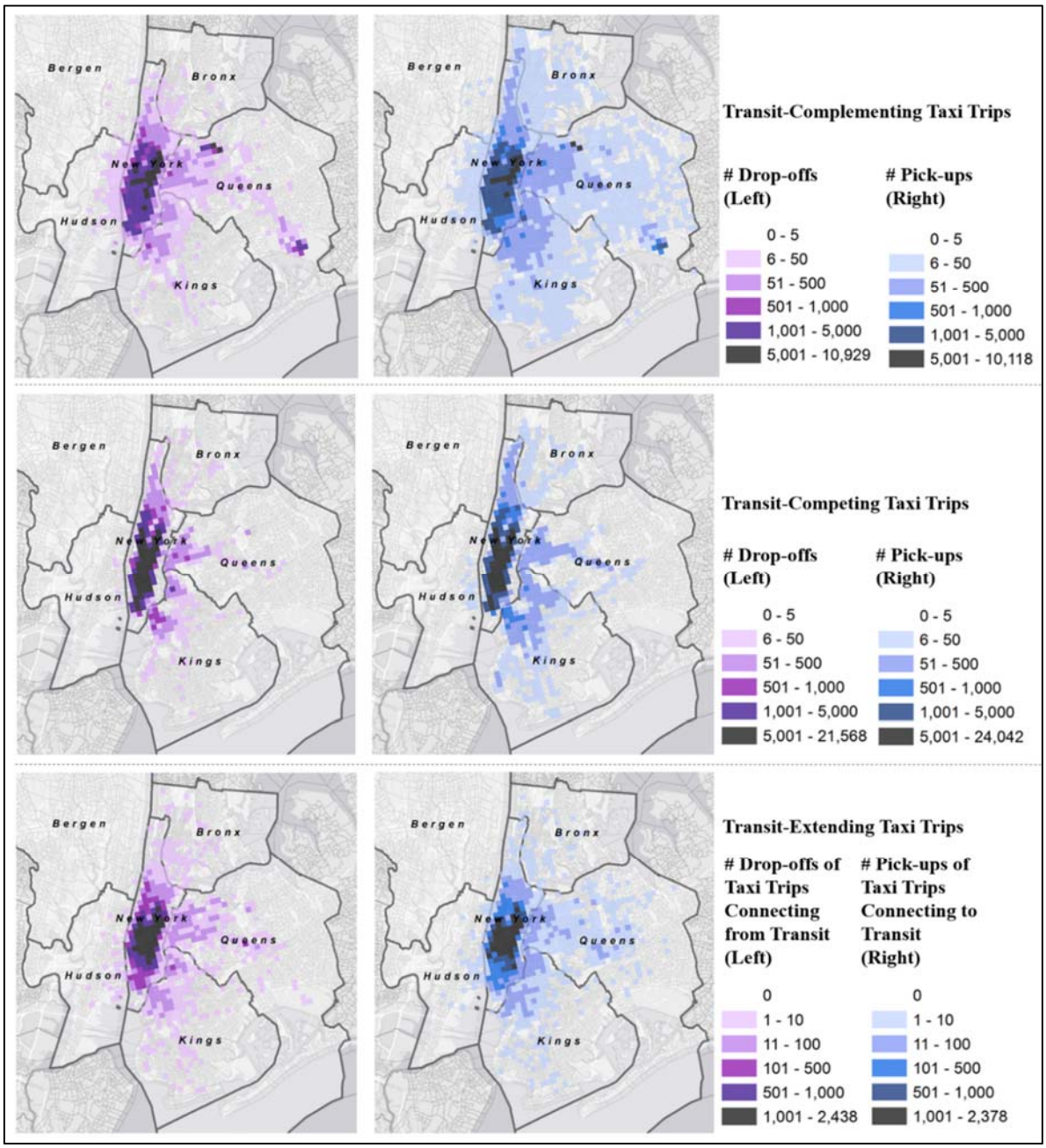

Figure 5. Spatial Distribution of the Origins/Destinations of the Three Categories of Taxi Trips

Spatial Distribution

The spatial distributions of the three types of taxi trips not only reflect the geographical locations of taxi trips, but also link the trips with built environment factors. Figure 5 maps the number of taxi pick-ups/drop-offs at the 200-meter by 200-meter grid level. Transit-complementing taxi trips present the most expansive scales, and the drop-offs of transit-complementing trips are more spread out than the pick-ups. Transit-competing trips follow a similar pattern to subway lines, which is pre-determined by trip classification. Transit-competing trips also show a more spread-out pattern of drop-off locations than pick-ups. The drop-offs and pick-ups of transit-extending trips do not vary much, and they concentrate mostly around subway stations. The finding that the drop-offs of taxi trips are more expansive than the pick-ups echoes with previous research that suggested the asymmetrical pattern of taxi trips.

Binary Logit Model Results

Descriptive statistics of the variables from the two binary logit models are summarized in Table 4. The variables included in the models were selected carefully to avoid collinearity, and different models were selected mainly by comparing their Akaike information criterion (AIC) and log-likelihood values.

Table 4. Summary of Descriptive Statistics of Variables Included in the Binary Logit Models

Continuous Variables	mean	s.d.	min	max
Log distance	2.71	3.08	0.10	82.70
Pick-up: Employment density	35,321	35,696	0	157,639
Pick-up; Employment-population balance	35	174	0	4,181
Pick-up: Median housing value	833,054	213,261	11,700	1,000,001
Pick-up: Poverty rate	0.05	0.09	0	1.00
Pick-up: Bus stop density	8	12	0	98
Pick-up: Bus line density	19,950	13,800	25	41,625
Drop-off: Employment density	35,037	37,090	0	157,639
Drop-off: Employment-population balance	30	177	0	4,181
Drop-off: Median housing value	814,953	224,865	11,700	1,000,001
Drop-off: Poverty rate	0.06	0.10	0	1.00
Drop-off: Bus stop density	7.75	11.25	0	98
Drop-off: Bus line density	25,400	17,375	25	52,975
Categorical Variables	**Number of 0**		**Number of 1**	
Morning peak	845,107		137,975	
Evening peak	835,127		147,955	
Late night	725,652		257,430	
Payment cash	434,766		548,316	
Rain dummy	904,125		78,957	
Pick-up land use: High residential	817,745		165,337	
Pick-up land use: Mixed use	678,605		304,477	
Pick-up land use: commercial	677,275		305,807	
Drop-off land use: High residential	785,384		197,698	
Drop-off land use: Mixed use	716,842		266,240	
Drop-off land use: commercial	681,155		301,927	

There are several interesting findings from the results of the two binary logit models, shown in Table 5. Regarding trip-level variables, the three time-related dummy variables, including morning peak, evening peak, and late night are very significant in both models, but have opposite signs. Transit-competing taxi trips tend to occur less during morning and evening peak hours, and more in late night, while transit-extending taxi trips take place more in peak hours and less in late night. This indicates that during rush hour, travelers tend to take transit to avoid roadway congestion, generating a higher proportion of transit-extending taxi trips. Travelers may also not want to take transit late at night, possibly due to longer headways and security concerns. As a result, transit-competing taxi trips happen more in late night, while transit-extending trips decrease significantly at night.

Table 5. Results of the Two Binary Logit Models

	Model 1- Dependent Variable: Transit-Competing Taxi Trips			Model 2- Dependent Variable: Transit-Extending Taxi Trips		
	Estimate	Std.Error	Pr(>|z|)	Estimate	Std.Error	Pr(>|z|)
(Intercept)	-1.34	0.02	***	-2.04	0.02	***
Trip level variables						
Morning peak	-0.16	0.01	***	0.14	0.01	***
Evening peak	-0.03	0.01	**	0.04	0.01	**
Late night	0.07	0.01	***	-0.14	0.01	***
Log distance	-0.21	0.00	***	-0.76	0.00	***
Payment cash	-0.02	0.01	***	0.03	0.01	***
Rain dummy				0.06	0.01	***
Pick-up locational variables						
Pick-up: Employment density	0.0015	0.00	***	-0.0004	0.00	***
Pick-up; Employment population balance	-0.0007	0.00	***			
Pick-up: Median housing value	0.000001	0.00	***			
Pick-up: Poverty rate	-0.25	0.03	***	0.32	0.04	***
Pick-up: Bus stop density	-5.75	0.25	***	3.75	0.25	***
Pick-up: Bus line density	-0.0025	0.00	***	0.0031	0.00	***
Pick-up land use: High residential	0.13	0.01	***	-0.20	0.01	***
Pick-up land use: Mixed use	0.20	0.01	***	-0.10	0.01	***
Pick-up land use: commercial	0.30	0.01	***			
Drop-off location variables						
Drop-off: Employment density	0.0017	0.00	***	-0.0007	0.00	***
Drop-off: Employment population balance	-0.0007	0.00	***			
Drop-off: Median housing value	0.000001	0.00	***			
Drop-off: Poverty rate	-0.09	0.03	**	0.21	0.04	***
Drop-off: Bus stop density	-5.5	0.25	***	5.25	0.25	***
Drop-off: Bus line density	-0.0025	0.00	***	0.0023	0.00	***
Drop-off land use: High residential	0.16	0.01	***	-0.22	0.01	***
Drop-off land use: Mixed use	0.25	0.01	***	-0.16	0.01	***
Drop-off land use: commercial	0.34	0.01	***	-0.09	0.01	***
Significance Codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1						

The dummy variable indicating whether it was raining (in that hour) when the taxi trip took place is also found to be positively correlated with transit-extending trips, indicating that there is higher rate of transit-extending trips when it is raining. However, this may not reflect travelers'

willingness to take transit when there is rainfall, but is likely to be a result of the limited supply of taxi during the rain. As Kamga, Yazici, and Singhal (2013) identified taxi drivers make more frequent and slightly shorter trips to increase their income when there is rainfall, and after reaching their income target, drivers may end their shift early, which might result in the perceived taxi shortage during prolonged rain. It is likely that the shortage of taxi supply during rainfall resulted in more transit trips and transit-extending trips. Temperature is not found to have a significant role in the two models.

Trip distance is a key variable that distinguishes the three types of taxi trips. The log transformed trip distance variable has a small negative coefficient in model 1 and a larger negative coefficient in model 2. This is consistent with the trip length distribution shown in Figure 4 and reflects that travelers who are farther than two miles from a transit station may not take a taxi to access it (or will not choose to take transit at all). Travelers' decision to take a taxi to access transit might be substantially related to the trip length and the cost which itself is nearly proportional to trip length. The data shows that about 90% of the transit-extending taxi trips are shorter than 2 miles and about 86.5% of them have a cost below $10, indicating that travelers may prefer taking a taxi to access transit when the trip is shorter than 2 miles and cost less than $10 given the current pricing of taxi service in NYC. Shorter travel distances tend to encourage travelers to choose taxi over transit, which might be due to the lower cost of taxi trips for shorter distances, less waiting time, and point-to-point travel convenience compared to transit.

Regarding the built environment factors associated with either the origin or destination of a taxi trip, employment density, median housing value, and poverty rate are found to be significantly associated with transit-competing trips. Employment density and median housing values are positively associated with transit-competing trips, and the poverty rate is negatively associated with transit-competing trips. This may be partially explained by the fact that subway and train stations are more likely to be located in the downtown area with higher employment density and land value. The positive correlation of transit-competing trips with high-density residential, commercial, and mixed-use land use also attest that the built environment characteristics of the origin or destination of transit-competing trips are to some extent predetermined by the characteristics of subway/train locations. In contrast, most of these built environment variables have the opposite signs for transit-extending trips. In Model 2, employment density has a negative sign, poverty rate has a positive sign, and the land use categories of high-density residential, mixed-use, and commercial (only at the destination end) have negative signs. This contrast suggests that different types of taxi trips are serving areas with very different economic and built environment characteristics. Transit-competing trips serve the areas where subway and train are easily accessible, which tend to be the highly developed downtown area with high employment density and housing value. Transit-extending trips, compared to both transit-competing and transit-complementing trips, are serving more areas with low employment and residential density, lower housing values, and higher poverty rates, which tend to be peripheral low-income areas. Interestingly, the variable of employment-population balance is found to have negative signs at both the origin and destination of transit-competing trips. It implies that better job-housing balance may decrease the probability of people taking a taxi to replace transit.

Bus stop density and bus line density are found to be negatively associated with transit-competing trips and positively associated with transit-extending trips. This demonstrates that transit-extending and transit-complementing trips are more likely to start from or end in areas with higher bus service density, compared to transit-competing trips. It implies that taxis are probably

replacing some bus trips that connect to subway or train stations, or substitute for bus trips in areas with poor subway/train service.

DISCUSSION AND CONCLUSIONS

This research analyzes taxi trips from an innovative perspective by quantifying the multifaceted relationship between the taxi and transit. The three distinct types of taxi trips, transit-competing, transit-complementing, and transit-extending trips, were examined by comparing their trip characteristics and their association with built environment factors. The analysis also includes an attempt to reveal a more complete demographic profile of taxi riders. This is challenging because linking taxi trip data with socio-demographic factors suffers from a lack of empirical data. Some important findings from the analysis are summarized and implications are discussed below.

The finding that approximately 59.5% of the taxi trips are serving disabled, low-income, elderly, retired or unemployed people reflects the significant paratransit role that taxi plays in our current transportation system. Although previous studies have provided some relevant evidence, our results also found a high proportion of taxi riders to be physically or economically vulnerable. This pattern might be partially due to the 'Access-A-Ride' program that NYC launched to reduce the cost of taxis for physically challenged people, but also reflects the strong dependency of the disadvantaged population on taxi services. Moreover, in the face of rapidly increasing ride-sourcing services and emerging travel modes that share similarities with taxis, it is important to consider how to improve physically or economically challenged people's access to such on-demand ride service. NYC has announced its goal of making 50% of its taxi fleet wheelchair accessible even though currently only about 1.8% of taxis are wheelchair accessible (Fitzsimmons 2015; Donohue 2013).

Regarding the relationship between the taxi and fixed-route mass transit, the analysis shows that different types of taxi trips are serving very different market segmentations. The analysis substantiated the fact that a substantial proportion of taxi trips may connect to/from transit stations. Though this type of trip categorization cannot be validated with empirical data, the much shorter distance of transit-extending trips reveals its intrinsic distinction from the other two types. We all know that the efficiency of a transit network not only depends on the transit system itself but also on how people can access the system. It is commonly accepted that transit walk sheds are about 0.5 miles from a station, meaning that people are willing to walk 0.5 miles to access transit. The multimodal connection between bike and transit has received more attention recently, mainly because biking can enlarge the catchment area of transit at low cost. However, biking has its own limitations as a transit access mode: biking is sensitive to weather, biking routes, and the number of people traveling; usable distance is limited by users' physical abilities, bike racks on buses and trains constrain capacity. By contrast, taxis do not have similar constraints.

Cost is the major constraint of taxis as a transit access mode compared with walking or biking. Based on our findings, the vast majority of transit-extending trips are less than two miles long with a cost less than ten dollars, indicating a common willingness-to-pay to access transit under current pricing. Though it is hard to predict the elasticity of taking a taxi to access transit, incentives to reduce the cost of transit-extending taxi trips should be considered as a strategy to improve the multimodal connection between the taxi and transit. Moreover, transit-extending taxi riders are found to use cash for payment more frequently than other taxi riders. King and Saldarriaga (2016) suggested that paying cash for taxi rides is associated with unbanked and low-income populations. Combining the payment methods of taxi and transit might promote taxi-transit integration. Given the substantial proportion of taxi trips taken to connect transit and the great

potential of the taxi or other on-demand ride services such as ride-sourcing or shared autonomous vehicles, it is worthwhile to consider such strategies and services to improve multimodal connections and encourage transit use.

Taxis share many similarities with ride-sourcing and the upcoming shared autonomous vehicles, as they do not require automobile ownership, serve point-to-point flexible routes, and are based on requests. Ride-sourcing is found to mostly replace taxi trips, but it also replaces some driving and transit trips (Rayle et al. 2016). Ridesharing and real-time information provision might be the key difference that distinguishes ride-sourcing and shared autonomous vehicles from traditional taxis. Some studies have explored the potential that dynamic ridesharing could have in affecting vehicle miles traveled (VMT) and urban land use patterns (Shaheen, Chan, and Micheaux 2015; Fagnant and Kockelman 2014; Zhang et al. 2015). Incorporating ridesharing into the taxi operation might be the next step to reduce some of the travel cost, especially for transit-extending taxi trips. Additionally, ridesharing largely depends on trip matching; trips starting or ending with transit stations are often easily matched with other trips, so strategies connecting ridesharing to transit can be effective in reducing VMT.

Although some built environment variables in our analysis, like employment density, median housing values, and land use, are correlated with some 'pre-determined' characteristics of transit stations' locations, the results of the models reveal significant differences in the areas that the three types of taxi trips are serving. Generally, transit-competing taxi trips are serving areas with great subway/train availability, which are mostly highly developed downtown areas with high land values. Transit-extending and transit-complementing trips are serving areas on the opposite side, which are more likely to be peripheral areas with low density and lower land values. However, this finding indicates the complementary role that taxis have in serving areas where transit does not operate efficiently. Moreover, the positive association between bus density and transit-extending trips indicates that taxis are probably replacing some bus trips in those areas. The different market segmentations that taxi serves are also related to the notion of 'captive' versus 'choice' users. Captive users are often defined as 'transit dependent' travelers who may not have many travel options, while choice users are referred to 'discretionary users' who can choose from several travel options (Giuliano 2005; Polzin, Chu, and Rey 2000; Jacques, Manaugh, and El-Geneidy 2013). It is hard to distinguish these two types of users in transit-complementing and transit-extending trips, meaning that it is hard to know whether a taxi rider chooses to take a taxi because of a lack of other options or simply because the taxi service maximizes the traveler's utility. However, compared to transit-competing trips, transit-complementing and transit-extending trips are more likely to be made by captive users, and accessibility analysis would be necessary to further identify the areas where people have to take the taxi because no other option is available.

This research broadens and deepens our understanding about taxi trips and offers a new perspective on the relationship between on-demand ride service and fixed-route public transportation in a multimodal transportation system. Taxis play a critical paratransit role in providing mobility to transit-dependent people, especially the physically and economically disadvantaged. They also support fixed-route mass transit and serve areas with insufficient transit service. Some specific findings reveal the direction that future policy and strategies might take to improve the multimodal connection between the taxi and transit. The research has several limitations. As mentioned before, the classification method cannot be validated and cannot fully encapsulate the transit-competing and transit-complementary nature of taxi trips. However, the vastly different trip characteristics of the three types of taxi trips buttress the validity of the

categorization. Another limitation is the lack of knowledge about the purposes of those taxi trips. We do not know to what extent the taxi trips are made by regular users, irregular users or by visitors/tourists. This emphasizes the need to collect taxi data that links taxi trip records with trip purposes and riders' socio-demographic information in the future.

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